# Investigating instrument resolution with McStas

ISIS 2025 McStas School

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# Resolution

## What do we mean by instrument resolution?

Two important concepts:

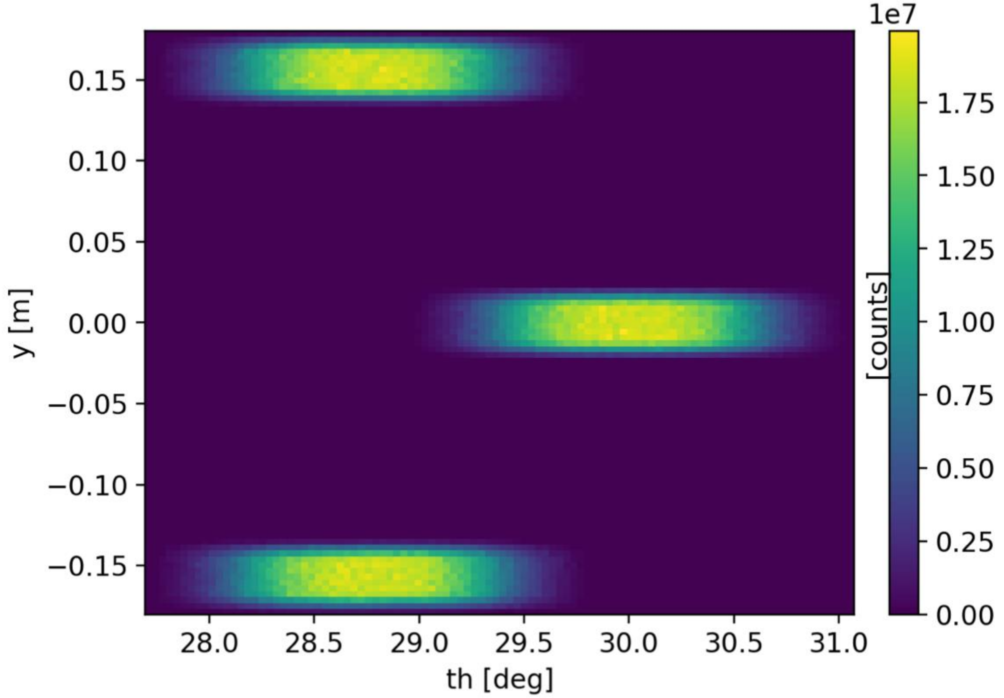
- **Point spread function**: Distribution on detector from specific **q**, E
  - Used to build expected detector response from known **q**, E distribution
- **Detector response**: Distribution of **q**, E measured by part of detector (pixel, time)
  - Used to build up map of **q**, E measured from detector response
- Depends on almost all aspects of instrument:
  - Source size/time structure, optics, sample size/shape, detector, …

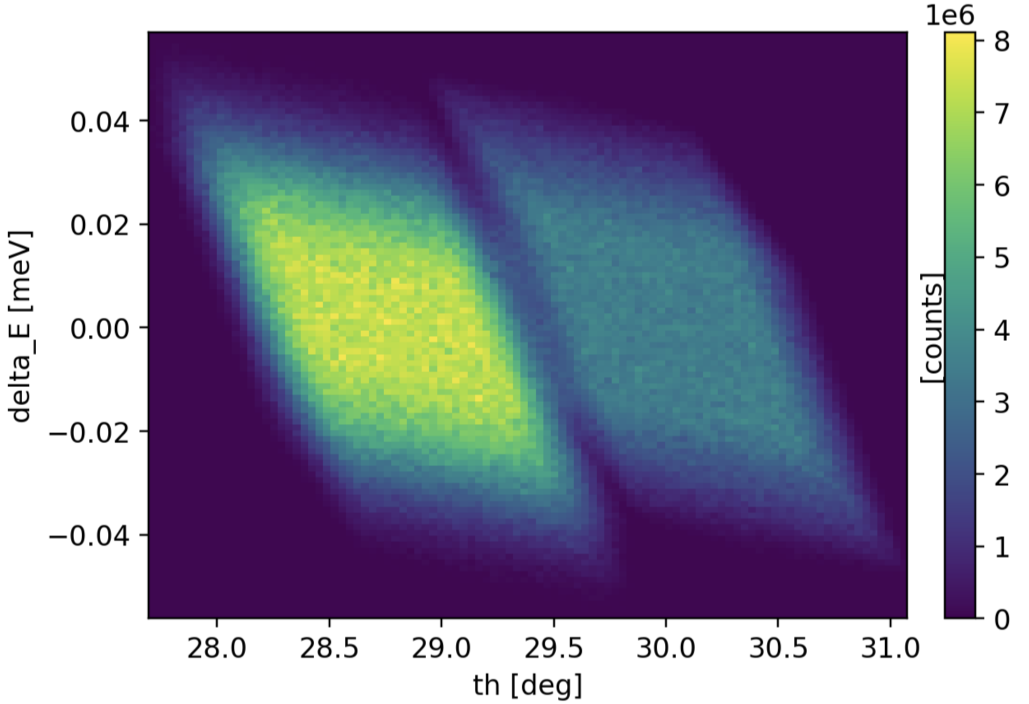
# Resolution

## Tools for point spread function

**Spot_sample** (in Contrib, by Garrett Granroth)

- Number of spots in **q**, E
- Probes the point spread function for all these spots in one simulation

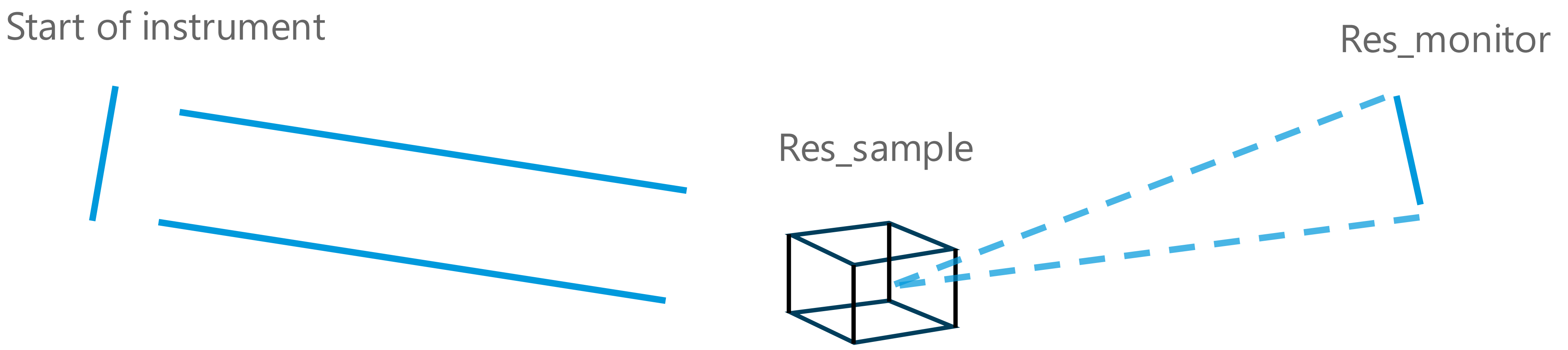


# Resolution

## Tools for detector response

- Continuous measurements **Res_sample.comp / Res_monitor.comp**




Sample sends rays towards monitor.
Selects random final energy in given interval.

Sample records **q**, E for each event
Monitor histograms those that reach it

# Resolution

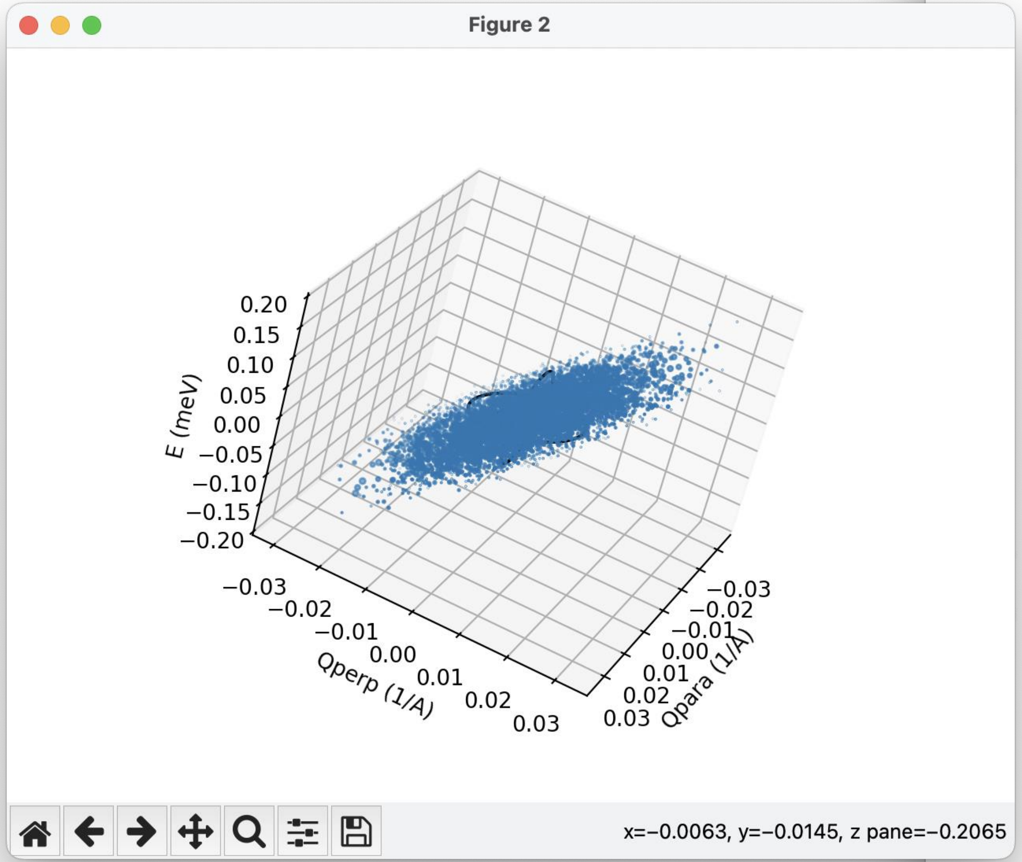
## Tools for detector response

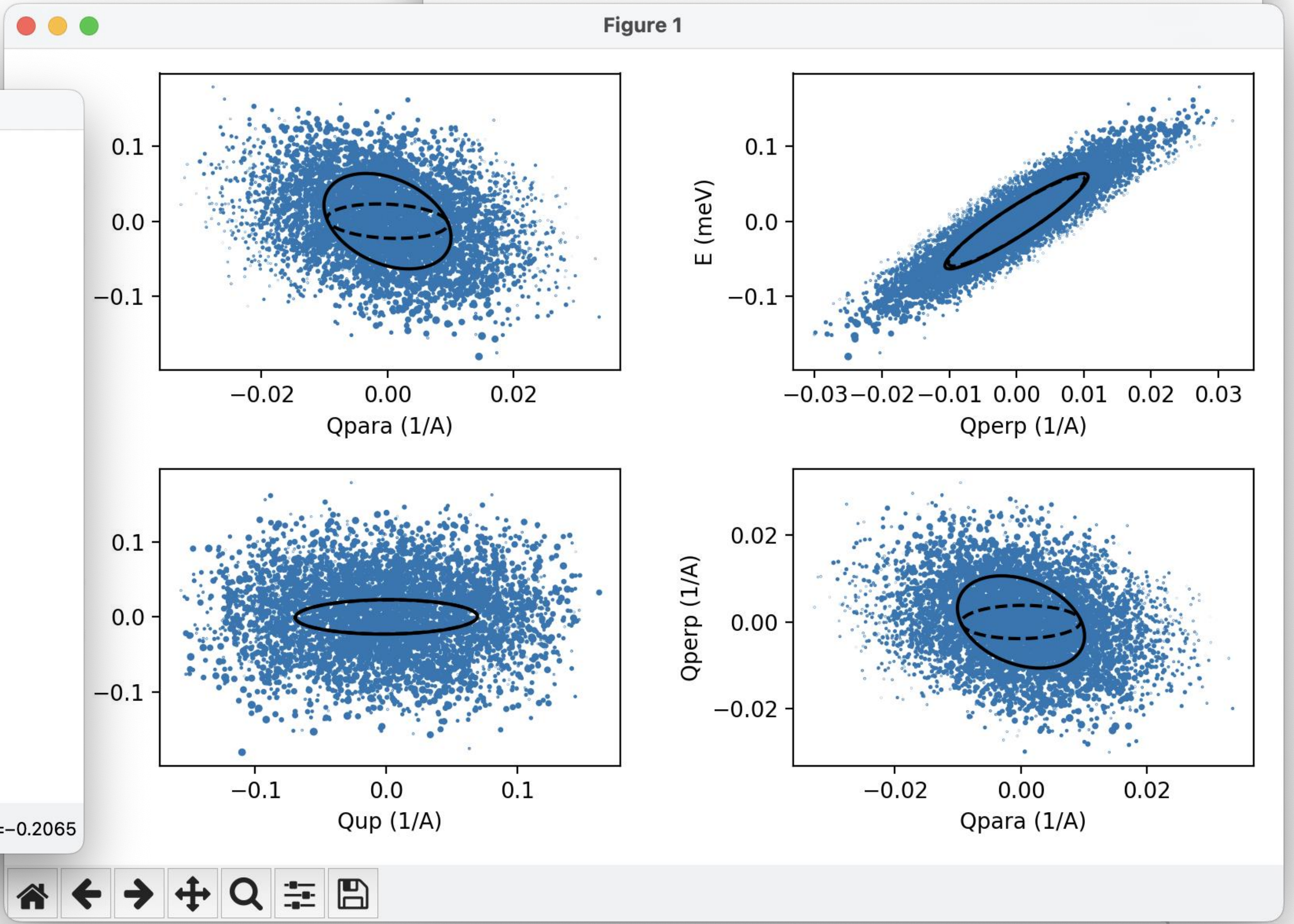
```
Test_TasReso_20251031_193910 — python3 ◂ mcresplot reso.dat — 92×61
(mcstas_fresh_2) madsbertelsen@CIN-101563 Test_TasReso_20251031_193910 % mcresplot reso.dat
This is a covariance matrix calculator using neutron events,
        doi: 10.5281/zenodo.4117437,
        written by T. Weber <tweber@ill.fr>, 30 March 2019.

Mean (Q, E) vector in lab system:
[-1.1213e+00  4.1128e-04  5.6444e-01 -2.7778e-03]

Covariance matrix in lab system:
[[ 9.3978e-05 -1.2078e-06  1.7440e-05  3.4412e-04]
 [-1.2078e-06  3.4789e-03  3.3121e-06  4.9043e-05]
 [ 1.7440e-05  3.3121e-06  6.1003e-05  3.3635e-04]
 [ 3.4412e-04  4.9043e-05  3.3635e-04  2.9168e-03]]
```

Terminal tool: mcresplot

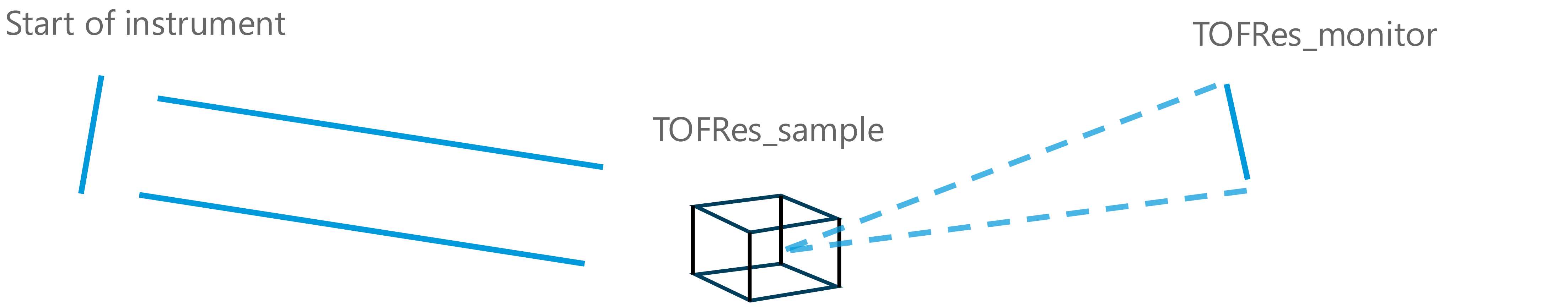


# Resolution

## Tools for detector response TOF case

- Time of flight measurements **TOFRes_sample.comp / TOFRes_monitor.comp**

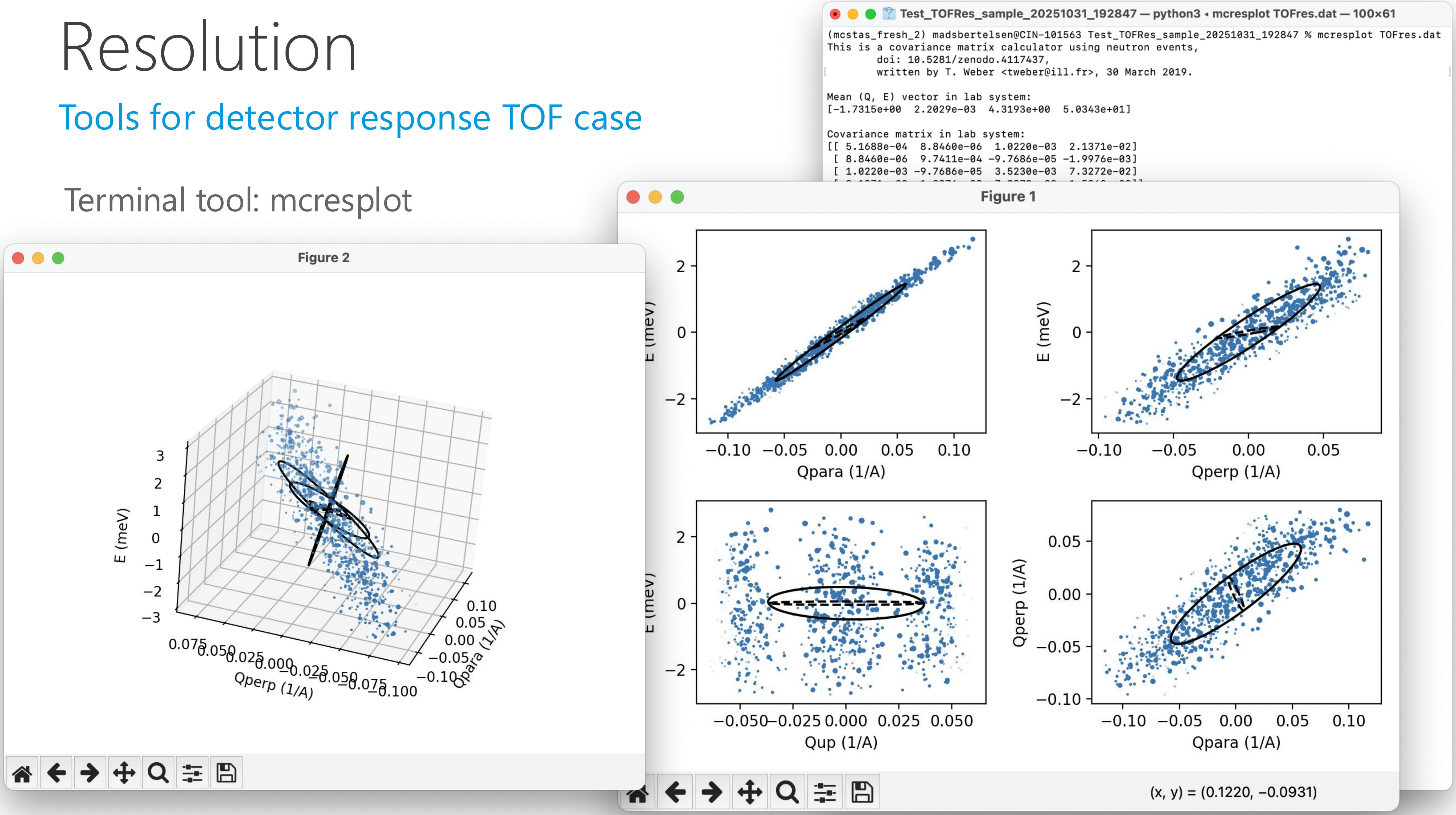



Sample records **q**, E for each event
Monitor histograms those that reach it

# Resolution

## Tools for detector response TOF case

Terminal tool: mcresplot



# Resolution

## Current situation

- Separate for detector response and point spread function
  - Point spread function can be determined for n spots at once with Spot_sample

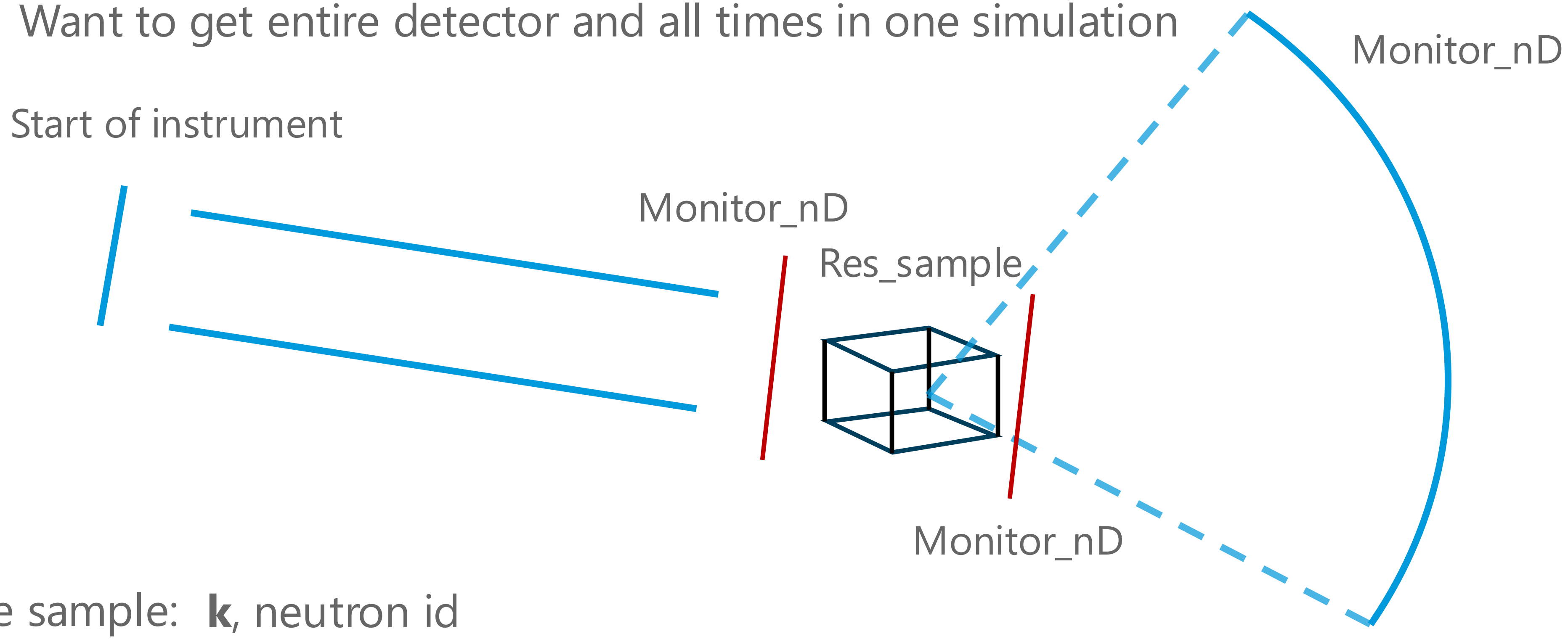
- For detector response
  - Existing solutions look at a single pixel (and time bin for TOF)
  - Getting resolution function for whole detector requires
    - Scan over instrument settings (incoming / final energy)
    - Either scan over all pixel positions or large number of components
    - For TOF instruments, scan over investigated time bin at given pixel

  - These components are admittedly designed for TAS / simple TOF instruments

# Resolution

## Method for full detector resolution

- Want to get both detector response and point spread function
- Want to get entire detector and all times in one simulation




Before sample: **k**, neutron id
After sample: **k**, neutron id
Detector: pixel position, time and neutron id

Sample sends rays towards monitor.
Selects random final energy in given interval.

# Resolution

## Method for full detector resolution

- For each event recorded in the detector we know
  - From the detector itself: pixel position, time and neutron id
  - From this id, we can look up: initial and final wavevector -> **q**, E

- With this information we can look at
  - Small subset of pixel / time positions -> Detector response for that area in **q**, E
  - Small subset of **q**, E -> Point spread function on detector

- This requires large dataset and code to handle data
- Still requires multiple simulations scanning over instrument settings